

Airline Passenger Dataset

1. Problem Definition

The objective of this project is to analyze **airline passenger and flight data** to understand passenger behavior, flight performance, ticket pricing, delays, cancellations, and loyalty patterns.

The analysis will help identify:

- Which airlines have more passengers and better flight status.
- The pattern of **on time, delayed, and cancelled flights**.
- Average ticket prices across different airlines and travel classes.
- Which routes and cities have higher passenger activity.
- The relationship between ticket price and travel class.
- Passenger booking preferences such as Website, Mobile App, and Travel Agent.
- Passenger loyalty based on loyalty points.
- Patterns in flight delays and cancellations.

Business Problem

Airlines need to understand passenger behaviour and flight performance to improve **customer satisfaction, operational efficiency, pricing strategies, and loyalty programs**.

This project uses historical passenger data to generate useful insights that can support these business decisions.

2. Dataset Selection

Dataset Name

Airline Passenger Dataset

Dataset Description

The dataset contains airline passenger and flight information. Each record represents a passenger's flight journey.

The dataset includes fields such as:

Category	Examples
Passenger Details	Passenger ID, Name, Age, Gender
Flight Details	Airline, Flight Number, Flight Type
Airport Details	Departure Airport, Arrival Airport
Location	Departure City, Arrival City
Travel Details	Travel Date, Departure Time, Arrival Time
Seat Details	Seat Number, Class
Financial Details	Ticket Price
Baggage	Baggage Weight
Flight Performance	Status, Delay Minutes
Booking	Booking Channel
Customer Preference	Meal Preference
Loyalty	Loyalty Points

Dataset Selection Reason

This dataset was selected because it contains both **categorical and numerical data**, making it suitable for data cleaning, exploratory data analysis, visualization, and business insight generation.

3. Business Understanding

Business Objective

The main business objective is to understand:

1. Passenger behaviour

- Passenger demographics
- Booking channel preferences
- Meal preferences
- Travel class preferences

2. Airline performance

- Flight status
- Delays
- Cancellations
- Passenger volume

3. Revenue and pricing

- Ticket price by airline
- Ticket price by class
- Ticket price by flight type

4. Customer loyalty

- Loyalty points by passenger
- Loyalty patterns across airlines and travel classes

5. Operational performance

- Delay patterns
- Frequently used routes
- On time versus delayed/cancelled flights

Key Business Questions

The analysis will answer questions such as:

- Which airline has the highest number of passengers?
- What percentage of flights are On Time, Delayed, and Cancelled?
- Which airline has the highest average ticket price?
- Which travel class has the highest ticket price?
- Which booking channel is most commonly used?
- Which routes have the highest passenger traffic?
- Which airlines experience more delays?
- Is there a relationship between loyalty points and travel class?
- How does ticket price vary between National and International flights?

4. Initial EDA

Before performing detailed analysis, the dataset will be explored to understand its structure, quality, and important patterns.

Initial EDA Steps

Step 1 – Dataset Structure

Check:

- Number of rows
- Number of columns
- Column names
- Data types

Step 2 – Missing Values

Check whether any columns contain missing/null values.

Important columns to check include:

- Age
- Airline
- Travel Date
- Ticket Price
- Status
- Delay Minutes
- Booking Channel
- Loyalty Points

Step 3 – Duplicate Records

Check for duplicate passenger or flight records and remove duplicates where appropriate.

Step 4 – Data Type Validation

Validate:

- Travel Date → Date
- Ticket Price → Numeric
- Baggage Weight → Numeric
- Delay Minutes → Numeric

- Loyalty Points → Numeric

The dataset also contains different date formats such as 01-02-2025 and 1/13/2025, so the Travel Date field should be standardized during cleaning.

Step 5 – Categorical Analysis

Identify unique values for:

- Gender
- Airline
- Flight Type
- Class
- Status
- Booking Channel
- Meal Preference

Step 6 – Numerical Analysis

Calculate:

- Mean
- Median
- Minimum
- Maximum

for numerical columns such as:

- Age
- Ticket Price
- Baggage Weight
- Delay Minutes
- Loyalty Points

5. Initial EDA – Expected Observations

From the initial exploration, we expect to examine:

Flight Status

Compare the number of:

- On Time
- Delayed
- Cancelled

flights.

Airline Distribution

Compare passenger records across airlines such as:

- IndiGo
- Air India
- Emirates
- Qatar Airways
- Singapore Airlines
- Lufthansa
- British Airways
- Delta

These airlines appear in the dataset.

Travel Class

Analyze passenger distribution among:

- Economy
- Premium Economy
- Business
- First

Booking Channel

Compare:

- Website
- Mobile App
- Travel Agent

Ticket Price

Analyze ticket prices based on:

- Airline
- Class
- Flight Type

Delay Analysis

Analyze:

- Number of delayed flights
 - Average delay minutes
 - Delay patterns by airline
 - Delay patterns by flight type
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Final TASK 1 Summary

Problem Definition:

Analyze airline passenger and flight data to understand passenger behavior, airline performance, ticket pricing, delays, cancellations, booking preferences, and loyalty patterns.

Dataset Selection:

An Airline Passenger Dataset containing passenger, flight, route, pricing, baggage, status, booking, and loyalty information was selected for the analysis.

Business Understanding:

The analysis aims to provide insights that can help airlines understand customers, monitor flight performance, improve operational efficiency, and support pricing and loyalty strategies.

Initial EDA:

The dataset will be examined for structure, missing values, duplicates, data types, categorical distributions, numerical statistics, flight status, ticket prices, booking channels, and delay patterns.

(<https://raw.githubusercontent.com/Naliniakshaya/Main-project-Airline-passenger-dataset/refs/heads/main/Air%20line%20Data%20sets.csv>)

```
import pandas as pd
```

```
df = pd.read_csv("https://raw.githubusercontent.com/Naliniakshaya/Main-project-Airline-passenger-dataset/refs/heads/main/Air%20line%20Data%20sets.csv")
```

```
df
```